

Government Architecture Readiness Checklist

UrbanPulse Smart City Deployment · MetroConnect · DSE ZG556

Checklist Completion

14 / 14 Items

PILLAR 1 — DATA SOVEREIGNTY

#01		All data must remain on city-owned on-premise servers UrbanPulse runs entirely on city-owned hardware . Kafka brokers, Flink, and Spark are deployed locally via Docker Compose. No cloud provider (AWS, Azure, GCP) is used. GPS positions, AQI readings, and electricity usage never leave the MetroConnect data centre. Verified by: no external bootstrap server in any producer config .	COMPLIANT
#02		Data residency — no cross-border transmission of citizen data Smart meter kWh readings and bus passenger occupancy data are personally linkable data under India's DPDP Act 2023. All Parquet files are written to local disk only . Kafka topic retention keeps data on-site for the mandated period (up to 365 days for smart meters). No Kafka MirrorMaker replication to any external cluster is configured.	COMPLIANT
#03		Network isolation — city data-centre firewall blocks inbound/outbound streaming Kafka listeners are bound to localhost / internal LAN only (no public IP). Producer <code>bootstrap.servers = localhost:29092</code> — unreachable from outside. Advisory API (FastAPI) exposes only read-only endpoints on the internal city network, not the public internet.	INFRA REQUIRED

PILLAR 2 — OPEN-SOURCE MANDATE

#04		100% open-source software stack — zero proprietary licences Full stack: Apache Kafka (Apache 2.0) · Apache Flink (Apache 2.0) · Apache Spark (Apache 2.0) · PostgreSQL (PostgreSQL Licence) · PostGIS (GPL 2.0) · InfluxDB OSS (MIT) · Grafana OSS (AGPL 3.0) · Python (PSF). No vendor lock-in. City IT team can modify, audit, and redistribute all components.	COMPLIANT
#05		All source code version-controlled and auditable in city Git repository Producers, Flink jobs, Spark scripts, and Docker Compose files are checked into a city-hosted Gitea instance . Any government auditor or ward officer can review the processing logic. No black-box SaaS analytics — every computation is transparent Python code.	COMPLIANT
#06		Data formats are open standards — JSON, Parquet, CSV (no proprietary formats) Kafka messages use plain JSON (RFC 8259). Batch output uses Apache Parquet (open columnar format). Councillor reports exported as PDF and CSV . Future systems can ingest UrbanPulse data without any vendor SDK.	COMPLIANT

PILLAR 3 — DISASTER RECOVERY (RPO < 15 MIN · RTO < 30 MIN)

#07		Kafka replication factor = 3 across all 3 brokers (RPO ≈ 0 for in-flight data) Every topic is created with replication-factor=3 and min.insync.replicas=2 . A single broker failure causes zero data loss — the surviving two brokers hold a complete replica. Producers use <code>acks=all</code> — a message is only acknowledged when all 3 replicas confirm write. RPO for Kafka stream = 0 ms under single-broker failure.	CONFIG
#08		Flink checkpointing every 5 minutes → RTO < 10 min for streaming jobs Flink is configured with RocksDB state backend and checkpoint interval = 5 minutes, saved to local disk. On restart after failure, Flink restores from the last checkpoint (max 5 min of data re-processed from Kafka). With Kafka topic retention ≥ 24 hours, replay is always possible. RTO for Flink < 10 min (5 min checkpoint gap + ~5 min restart time).	CONFIG
#09		Nightly Parquet snapshots taken — historical data survives server failure A nightly cron job (<code>rsync</code>) copies Parquet output directories to a secondary city server . If the primary server fails, councillor reports can be re-generated from the secondary copy within 30 minutes. Parquet files are immutable — no risk of partial writes corrupting the archive.	INFRA REQUIRED
#10		Dead-letter queue prevents data loss from validation failures Messages that fail validation (null AQI, impossible GPS coords, out-of-range voltage) are routed to urbanpulse.dlq topic — never silently dropped. A 5-minute DLQ error-type distribution report is generated automatically. DLQ topic retention = 7 days — enough time for operations team to investigate and replay if needed.	COMPLIANT

PILLAR 4 — DASHBOARD ACCESSIBILITY FOR WARD OFFICERS

- #11

Grafana dashboards require no SQL or technical knowledge — pre-built panels only

CONFIG

Ward officers access a **read-only Grafana dashboard** with pre-built panels: live AQI gauge, bus fleet heatmap, ward energy consumption tile. No query language required. All panels load automatically. Dashboard text uses **plain English, not technical jargon** (e.g. "Air Quality Level" not "PM2.5 µg/m³ rolling 10-min avg").
- #12

Councillor reports available as auto-generated PDF — no system access required

CONFIG

Weekly ward energy summaries and monthly AQI reports are auto-emailed as **PDF attachments** generated by the Spark batch job. Councillors receive reports without logging into any system. PDF uses plain tables (no charts) so it prints cleanly on standard A4 paper — accessible to officers in areas with poor monitor availability.

PILLAR 5 — SECURITY & COMPLIANCE

- #13

Kafka SASL authentication — only authorised producers/consumers can connect

INFRA REQUIRED

Kafka brokers are configured with **SASL/PLAIN** authentication (username + password per service). Separate credentials for: air quality producer, bus GPS producer, Flink job, Spark job, advisory API consumer. Prevents rogue sensors or unauthorised services from injecting fake data into the pipeline.
- #14

Audit log — all pipeline configuration changes are logged with timestamp and operator ID

INFRA REQUIRED

Any change to Kafka topic configuration, Flink job parameters, or Spark window sizes is recorded in a **city audit log**. Provides accountability trail for state government inspections. Log stored in append-only PostgreSQL table — operators cannot modify past entries.

Checklist Summary — 14 Items Across 5 Pillars

7

Compliant / Implemented

4

Config / Code Required

4

Infrastructure Required

0

Gaps / Non-compliant